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BHOPAL

Project Title: AIML Portfolio

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INTRODUCTION

This Student Study Planner is a Python-based command-line tool designed to help students track and prioritize their academic workload. By calculating the number of days remaining between today and a task's deadline, the script automatically sorts assignments by urgency and assigns priority tags like "DO THIS NOW" or "OVERDUE." It serves as a simple yet effective utility for managing time and staying organized during busy academic terms.

Project Report: Student Study Planner

1. Project Overview

The Study Planner is a utility script that calculates the time remaining for academic tasks. It allows users to input multiple assignments and automatically sorts them by urgency, providing visual "tags" to indicate which tasks require immediate attention.

2. Key Features

- **Dynamic Task Entry:** Users can input an unlimited number of assignments until they type "done."
- **Automatic Priority Sorting:** The script uses a sorting algorithm to place the most urgent tasks at the top of the list.
- **Status Tagging:** Based on the days remaining, the script assigns one of three statuses:
 - **!!! OVERDUE !!!:** For tasks past their deadline.
 - **DO THIS NOW:** For tasks due within 48 hours.
 - **Relax:** For tasks due further in the future.
- **Error Handling:** Includes a try-except block to prevent the program from crashing if an invalid date format is entered.

3. Technical Logic

- **Library:** Utilizes the datetime module for real-time date calculations.

- **Data Structure:** Tasks are stored in a **List of Dictionaries**, making it easy to manage multiple attributes (name, date, and calculated days) for each entry.
- **Sorting:** Uses a lambda function to sort the list based on the integer value of `days_left`.

4. Code Refinement Suggestions

While the current script is functional, the following improvements would enhance usability:

- **Input Validation:** The current except block defaults to 999 days. Providing a specific error message (e.g., "Invalid format, please use YYYY-MM-DD") would be more user-friendly.
- **Persistent Storage:** Currently, data is lost when the script closes. Adding a feature to save tasks to a .txt or .json file would allow for long-term use.
- **Correction in Output:** There is a small typo in the final print f-string where `t['date']` is printed twice instead of `t['days']`.

5. Conclusion

The Study Planner is an effective "Minimum Viable Product" (MVP) for task management. It successfully automates the mental effort of calculating deadlines, allowing students to focus on their actual coursework.

